Edge Computing Migration

VMware & Cloud to KVM/KubeVirt at the Edge

How enterprises are eliminating VMware licensing at remote sites, repatriating cloud workloads, and deploying lightweight KVM + K3s infrastructure — saving 80%+ on edge compute costs.

hyper2kvm + hypersdk | Apache 2.0 Open Source

The Edge Challenge

Running VMware at the edge was expensive before Broadcom. Now it's unsustainable.

VMware at Every Site = Multiplied Cost

Each remote site needs vSphere licenses. 50 sites x $10K/year = $500K just for hypervisor licensing. Broadcom increases make this 2-5x worse.

Cloud Egress Costs Explode

Edge workloads generate data locally but cloud-hosted VMs pay egress fees. $0.09/GB adds up to thousands monthly for video, IoT, and analytics workloads.

Latency Kills Cloud Edge

Manufacturing, retail, and healthcare need sub-10ms response times. Cloud round-trip adds 20-100ms. Local KVM eliminates network dependency entirely.

Remote Sites = Limited Resources

Edge locations have constrained hardware, unreliable networking, and no on-site IT. Heavy hypervisors like vSphere waste precious compute on management overhead.

$500K+
Annual VMware licensing for 50 edge sites

80%
Cost reduction with KVM + K3s at edge

10x
Less management overhead per site

Cloud Repatriation

Why enterprises are bringing workloads back from AWS, Azure, and GCP — and how hyper2kvm makes it painless.

Cloud (Before)

- $8,000-15,000/month per workload
- Egress fees on every data transfer
- Vendor lock-in (AMI, managed services)
- Compliance issues (data residency)
- Latency for edge-adjacent workloads
- Unpredictable monthly bills

On-Prem KVM (After)

- $500-2,000/month (hardware amortized)
- Zero egress fees — data stays local
- Open formats (qcow2, libvirt XML)
- Full data sovereignty
- Sub-millisecond local access
- Fixed, predictable costs

Cost Comparison: 20 VMs Over 3 Years

Cost Item	AWS/Azure	On-Prem KVM	Savings
Compute (VMs)	$720,000	$60,000 (hardware)	$660,000
Storage	$180,000	$15,000 (local SSD)	$165,000
Network / Egress	$90,000	$0	$90,000
Licensing	$0 (included)	$0 (KVM is free)	$0
Migration (one-time)	N/A	$5,000	-$5,000
Operations / Support	$36,000	$30,000	$6,000
3-Year Total	**$1,026,000**	**$110,000**	**$916,000**

> "We moved 20 VMs from AWS to on-prem KVM and saved $900K over 3 years. The migration took 2 weeks with hyper2kvm."

Edge Architecture

Lightweight, zero-license infrastructure for every remote site.

VMware / Cloud VMs → hyper2kvm export → K3s + KubeVirt → Edge Site

K3s (Lightweight K8s)

Single binary, 512MB RAM, runs on ARM or x86. Perfect for edge hardware. CNCF certified Kubernetes — same APIs as full clusters.

KubeVirt on K3s

Run VMs as Kubernetes pods. Manage 50 edge sites from one central cluster. GitOps deployment with Flux/ArgoCD.

hyper2kvm Convert

Export from vSphere/cloud, convert to qcow2, upload to PVC. One YAML config per site. Fully automated.

Edge Stack (Per Site)

Layer	Technology	Cost
Hardware	Intel NUC / Dell Edge / HPE EL	$1,000-3,000 one-time
OS	Fedora / RHEL / Ubuntu	$0 (or $500/yr RHEL)
Kubernetes	K3s	$0
VM Runtime	KubeVirt + KVM	$0
Storage	local-path / Longhorn	$0
Migration Tool	hyper2kvm	$0
Total per site		**$1,000-3,500**

Compare: VMware vSphere Essentials at one edge site = $6,000-15,000/year in licensing alone. K3s + KubeVirt = $0/year in software costs.

Industry Use Cases

Edge migration scenarios across sectors.

Manufacturing / Factory Floor

Challenge: SCADA/MES VMs on VMware at 30 plants. $300K/yr in licenses.
Solution: Migrate to K3s + KubeVirt on ruggedized edge servers. GitOps manages all plants from HQ.
Savings: $280K/yr. Faster updates, zero license audits.

Retail / Point of Sale

Challenge: POS and inventory VMs in 200 stores on aging ESXi hosts.
Solution: Mini PC with K3s at each store. hyper2kvm converts VMDKs centrally, ship qcow2 images.
Savings: $1.2M/yr. Standardized deployments, remote management.

Healthcare / Clinics

Challenge: EMR and imaging VMs at 50 clinics. HIPAA requires local data. Cloud egress costs $40K/mo.
Solution: Repatriate from Azure to local KVM. Data never leaves the clinic. HIPAA compliant by design.
Savings: $480K/yr in cloud costs. Full data sovereignty.

Energy / Remote Sites

Challenge: Monitoring VMs at 100 substations. Unreliable WAN. VMware licenses at each site.
Solution: K3s on ARM-based edge gateways. VMs run autonomously, sync when connected.
Savings: $800K/yr. Works offline, self-healing with K8s.

Construction / Temporary Sites

Challenge: Project management VMs deployed per-site, 6-18 month lifecycle. VMware overkill.
Solution: Portable edge box with K3s. Deploy in minutes via USB. Tear down when done. Reuse hardware.
Savings: 90% vs cloud. Zero recurring license cost.

Telecom / Cell Towers

Challenge: vRAN and core network VMs at thousands of sites. NFV licensing spiral.
Solution: Migrate VNFs to KubeVirt on bare-metal K8s. CNF-ready path forward.
Savings: 60-80% on NFV licensing. Path to cloud-native network functions.

Multi-Cloud Exit Strategy

hypersdk supports 10 cloud providers. Export from anywhere, land on KVM.

AWS EC2 → hypersdk → On-Prem KVM

Azure VMs → hypersdk → Edge K3s

GCP / OCI → hypersdk → KubeVirt

Cloud Exit Cost Savings

Source	Typical Monthly Cost	On-Prem Equivalent	Monthly Savings
AWS EC2 (m5.2xlarge x 10)	$5,500	$800 (amortized)	$4,700
Azure D8s_v3 x 10	$4,800	$800	$4,000
GCP n2-standard-8 x 10	$4,200	$800	$3,400
S3/Blob storage (10TB)	$2,300	$200 (local NVMe)	$2,100
Data egress (5TB/mo)	$450	$0	$450
Total (10 VMs)	**$17,250/mo**	**$2,600/mo**	**$14,650/mo**

Annual savings for 10 VMs: $175,800. Payback period on hardware: 2-4 months.

GitOps Edge Management

Manage 100+ edge sites from one Git repo. No SSH, no manual intervention.

Fleet Management

ArgoCD or Flux watches a Git repo. Push a VM manifest change, it deploys to all 50 sites automatically. Rollback by reverting a commit.

Declarative VMs

VMs defined as KubeVirt YAML in Git. Version controlled. Auditable. Reviewable. Same workflow as container deployments.

Zero-Touch Provisioning

New edge site: plug in hardware, boot PXE, K3s auto-joins fleet. ArgoCD deploys VMs. No SSH needed. No on-site IT required.

Central Observability

Prometheus + Grafana federated across all sites. Single dashboard for CPU, memory, disk across entire fleet. Alerting on any anomaly.

Example: Deploy VM to All Edge Sites

```
# git push this to deploy a VM to every edge site:
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: inventory-app
  namespace: production
spec:
  running: true
  template:
    spec:
      domain:
        resources:
          requests:
            memory: 2Gi
            cpu: "2"
        devices:
          disks:
```

```yaml
      - name: rootdisk
        disk:
          bus: virtio
    volumes:
    - name: rootdisk
      persistentVolumeClaim:
        claimName: inventory-app-disk
```

Edge Migration Automation

Convert once at HQ, deploy to every edge site automatically.

1. Export from vSphere → 2. Convert (hyper2kvm) → 3. Push to Registry → 4. GitOps Deploy

Centralized Conversion, Distributed Deployment

```
# Step 1: Export + Convert at HQ
sudo h2kvmctl --config edge-vm.yaml

# edge-vm.yaml
cmd: vsphere
vcenter: vcenter.hq.corp
vs_action: export_vm
vs_vm: factory-scada-01
output_dir: ./edge-images
out_format: qcow2
flatten: true
regen_initramfs: true

# Step 2: Upload to container registry as DataVolume source
virtctl image-upload dv factory-scada-disk \
  --size=10Gi \
  --image-path=./edge-images/factory-scada-01.qcow2 \
  --uploadproxy-url=https://cdi-proxy.corp

# Step 3: ArgoCD syncs KubeVirt VM manifest to all edge K3s clusters
# (automatic — just push the YAML to Git)
```

Migration Time per VM

- Export from vSphere: 5-30 min (depends on disk size)
- Convert + fix: 2-10 min
- Upload to registry: 5-15 min

Scale

- Convert 10-50 VMs per day at HQ
- Deploy to 100+ sites simultaneously
- Queue-based processing, no babysitting

- GitOps deploy to all sites: < 1 min
- **Total: ~15-60 min per VM**

- Per-VM boot verification
- **Full fleet migrated in 1-2 weeks**

Total Cost of Ownership

Side-by-side comparison across all approaches.

50 Edge Sites, 5 VMs Each (250 VMs Total), 3-Year TCO

Cost Category	VMware vSphere	Cloud (AWS/Azure)	K3s + KubeVirt
Software Licensing	$1,500,000	$0 (included)	$0
Compute	$300,000 (hw)	$4,500,000	$300,000 (hw)
Storage	$150,000	$900,000	$150,000
Network / Egress	$50,000	$540,000	$50,000
Management Platform	$200,000 (vCenter)	$0	$0 (GitOps)
Operations Staff	$450,000	$300,000	$250,000
Migration (one-time)	N/A	$50,000	$30,000
3-Year Total	**$2,650,000**	**$6,290,000**	**$780,000**
Savings vs VMware	—	**-$3,640,000**	**$1,870,000**
Savings vs Cloud	$3,640,000	—	**$5,510,000**

$1.87M
Saved vs VMware over 3 years

$5.5M
Saved vs Cloud over 3 years

70%
TCO reduction vs VMware

88%
TCO reduction vs Cloud

Next Steps

Start your edge migration journey today.

Edge Site Assessment

Inventory VMs at all edge sites. Classify by workload type (always-on, batch, seasonal). Map hardware capabilities. Identify quick wins.

Single-Site Pilot

Pick one edge site. Deploy K3s + KubeVirt. Migrate 3-5 VMs with hyper2kvm. Validate: boot, network, application, performance.

Fleet Rollout Plan

Based on pilot results, create phased rollout. Batch 1: 5 sites (low risk). Batch 2: 15 sites. Batch 3: remaining. 8-12 week timeline.

Decommission Legacy

Cancel VMware licenses site-by-site as migration completes. Repurpose or retire old hardware. Document savings for leadership report.

Save $1.87M Over 3 Years

Replace VMware at the edge with open source KVM + KubeVirt.
Zero license fees. GitOps management. Production-proven at scale.

hyper2kvm + hypersdk — github.com/ssahani/hyper2kvm